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Cracking finance: Why the quantum threat could arrive before the reward

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In Summary

- **Most technologies create value before they create problems, but quantum computing may break that pattern.** Imagine two clocks ticking side by side: one measures the time until quantum computers become economically useful, the other the time until they can break the cryptography that underpins the financial system. Which clock strikes first is a genuinely open race.
- **The upside potential is real but narrowly concentrated around derivative pricing and other complex financial simulations. Beyond these areas, its impact on the financial industry is likely limited.** Quantum computing accelerates Monte Carlo methods behind financial simulations, while it shows no consistent advantage yet over improving classic methods for portfolio optimization and financial machine learning. Insurers are well positioned as simulation and optimization are at the heart of liability valuation, risk pricing, capital modelling and asset-liability management. Quantum money could use uncopyable quantum states to authenticate value and prevent counterfeiting, although commercial deployment remains remote. The rewards remain theoretical for now, threatened by error-correction overhead and a rising classical benchmark, until large machines exist. But quantum-inspired methods deliver operational value on classical hardware today, and the talent, tooling and know-how built along the way will carry over.
- **Q-day could come sooner than we think: data harvested today can be decrypted later.** The recipe for breaking today's encryption has existed since 1994 and waits only for the machine, which is also the smaller one, needing roughly 1,250–1,450 logical qubits vs around 4,700–7,500 for leading financial applications. Health records, classified government information and sensitive financial data must stay confidential for decades, but experts put the probability of Q-day, the point at which a cryptographically relevant quantum computer exists, arriving within ten years at 28–49%. Defensive standards have existed since 2024, and regulators have set dates: The US National Institute of Standards and Technology (NIST) will phase out today's algorithms after 2030, the EU requires critical financial infrastructure to migrate by the same year.
- **Digital assets make the vulnerability visible: USD400bn of Bitcoin is already at risk, and that is just the tip of the iceberg.** By spring 2026, roughly 6m bitcoins, 30% of supply, sat behind exposed public keys, and Ethereum extends the exposure to stablecoins, bridges and tokenized assets. Spillover to mainstream finance would be systemic: The BIS puts losses above 1% of GDP over 15–20 years, while a severe Hudson Institute scenario reaches up to USD3.3trn in GDP losses (3% of global GDP) over 6 quarters.
- **Both private and public sectors are visibly chasing the reward, while concerns about the threat play out mostly behind closed doors.** Investors bet mostly on the hardware: about 97% of each dollar funds the machine and under 1% the applications, on an estimated USD12.6bn raised by quantum startups in 2025. Public efforts reveal a geographic imbalance: The US prepares for reward and threat via parallel executive presidential orders, while Europe, which draws just 5% of private quantum investment versus over half for the US, pursues its own response through fragmented policy instruments, risking a repeat of its AI experience as adopter and regulator of a technology it does not own. Preparation for the threat is purely an internal IT budgeting decision, competing resources with AI, and only about 35% of large organizations completed a cryptographic inventory. Institutions need crypto-agile architectures, prioritized inventories, migration roadmaps and tested fallbacks before the clock reaches zero: preparing for only one clock is the wrong strategy, regardless of which strikes first.

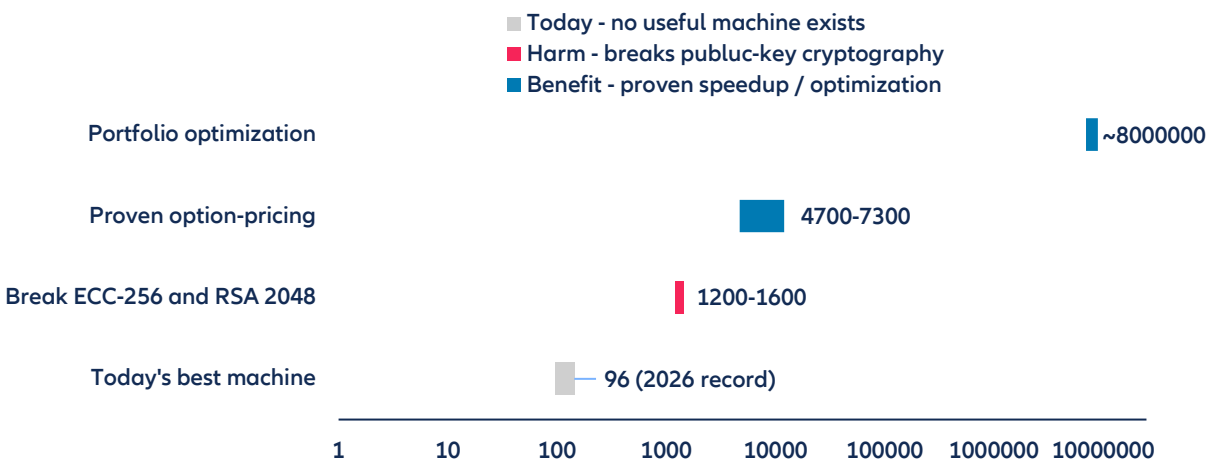
Two clocks ticking at different speeds

Most technologies create value before they create problems. The automobile delivered mobility before congestion became a policy concern; the internet created a digital economy before it created cybercrime. Even when the benefits arrived slowly – electrification took decades to lift factory productivity, and computers were once, as Robert Solow famously observed, visible "everywhere but in the productivity statistics" – the direction was the same: the rewards came first, and the risks followed. Quantum computing may break that pattern. Imagine two clocks ticking side by side. One measures the time until quantum computers become economically useful; the other, the time until they can break the cryptography that underpins the financial system. There is a credible case that the threat clock is running at least as fast as the reward clock, and that institutions, drawn to the visible upside, are quietly preparing for only one of the two.

To see why the two clocks can run at different speeds, start with the machine itself. A quantum computer is not simply a faster version of a conventional computer, nor does it "try every answer at once". It is a different class of machine that can solve a narrow set of mathematical problems in ways that remain intractable for conventional systems. Its basic unit is the qubit, the quantum counterpart of the classical bit. Whereas a conventional bit must be either 0 or 1, a qubit can occupy a quantum state that allows information to be processed differently. The difficulty is that qubits are fragile. The physical qubits built in today's laboratories are sufficiently error-prone that long computations quickly dissolve into noise. Useful large-scale quantum computing therefore depends on error correction: bundling large numbers of physical qubits into a much smaller number of stable logical qubits. Logical qubits, rather than raw physical-qubit counts, are the more meaningful measure of what a machine can actually compute. Even then, quantum advantage is highly specific. Most operations performed by banks, insurers and asset managers would see little benefit from quantum hardware. Yet one of the problems for which quantum computers are exceptionally well suited is factoring and discrete logarithms, the mathematical foundations of much of today's public-key cryptography.

On paper, and only against the most space minimized estimates, the code-breaking machine can be the smaller one. Attacking the elliptic-curve cryptography behind a Bitcoin key has been estimated to require roughly 1250–1,450 logical qubits, depending on circuit design and the trade-off between qubit count and runtime. The lowest estimates for breaking RSA-2048, a distinct problem from the elliptic-curve case, have fallen to a similar order of magnitude as architectures and algorithms have improved, though the standard single-shot figure stays several times higher. By comparison, a leading fault-tolerant proposal for pricing and risk using amplitude estimation requires roughly 4,700–7,500 logical qubits, while a general and economically relevant portfolio-optimization advantage may require far more resources and has not yet been demonstrated (Figure 1).

Figure 1: Estimated number of logical qubits required for different tasks



Sources: Babbush et al. (2026), Gidney (2025), Gong et al. (2026), Dalzell et al. (2023), Allianz Research. Note: The estimation of the number of logical qubits for portfolio optimization comes from a study using a method called the quantum interior point method (QIPM). Under this method, the number of qubits scales as $800n^2$, where n is the number of assets in the portfolio.

A simple count of logical qubits, however, does not settle the matter. Gate counts, fidelities and circuit depth matter as much as qubits and the deeper asymmetry is competitive rather than technical. For breaking cryptography, a quantum computer is essentially the only tool that can do the job; there is no classical shortcut, so the code-breaking machine only must exist. For finance, quantum must compete: classical hardware and algorithms already perform these calculations and keep getting better, so even a machine that is smaller on paper must clear a far higher bar to be worthwhile. Cryptographic attacks thus appear to be among the smallest clearly specified fault-tolerant applications, while the best-developed financial use cases demand larger and faster machines and must still beat a moving classical benchmark that the attacker never faces. The same asymmetry shapes the economics of building the machines: money spent chasing a financial edge competes against cheaper classical improvements, whereas money spent on defense buys protection against something only quantum can do.

The decisive asymmetry, though, is about recipes rather than machines. Shor's algorithm, the quantum procedure published by Peter Shor in 1994 that efficiently factors large numbers and solves the related discrete-logarithm problem on which public-key cryptography rests, has existed already for three decades. It is proven, published and has a clearly defined target: given sufficiently capable fault-tolerant hardware, it can recover the private keys behind the public-key cryptography, which authenticates and encrypts nearly every digital exchange, whether a bank transfer, a private message or a social network post. The remaining uncertainty is about the time when a machine with scale, speed and error correction sufficient to run the algorithm will exist.

Financial quantum algorithms are not absent, but their route to economic value is far less settled. Quantum amplitude estimation offers a firm mathematical basis for accelerating Monte Carlo calculations, while other approaches target optimization, machine learning and linear algebra. Yet for amplitude estimation the potential gain is quadratic rather than exponential, and for optimization and machine learning many claimed advantages remain heuristic or highly problem specific. The destructive use case already has its recipe and waits mainly for the machine. Productive uses need two things at once: the very hardware the adversary is also waiting for, and financial algorithms that still have to be found, implemented and shown to outperform strong classical methods. However, the two clocks compete on different dimensions, both worth exploring, and it is hard to say which clock will strike first.

The reward clock

The strongest case for quantum computing in finance starts from a simple observation: the industry already devotes enormous computing capacity to simulating uncertainty. Derivative pricing, market-risk measurement, economic-capital calculations and stress testing rely heavily on Monte Carlo methods – repeated simulations of thousands or millions of possible future outcomes. Quantum amplitude estimation can, in principle, achieve a given level of statistical accuracy with fewer evaluations than classical Monte Carlo. The potential result is not a wholly new calculation but the ability to perform existing calculations faster, more frequently or at greater resolution: compressing overnight risk runs, refreshing capital positions more often and evaluating a broader range of scenarios. Because Monte Carlo methods are used for a variety of applications, like pricing complex derivatives, calculating Value-at-Risk and expected shortfall, estimating counterparty exposure or measuring economic capital, the industry is unusually well aligned with one of the few quantum applications supported by a firm mathematical speedup.

The gain should nevertheless not be overstated. A quadratic acceleration can be valuable, especially for deeply nested simulations, but the moving classical benchmark takes concrete form here: specialized chips, GPUs, distributed cloud infrastructure, variance reduction and continually improving classical algorithms. A commercially useful quantum system must outperform them after incorporating error correction, data loading, state preparation and the cost of the full workflow. This is the single most important test any financial quantum application has to pass, and it is why even a genuine speedup on paper may not translate into a genuine advantage in practice.

Insurers sit at a competitive intersection: the one quantum advantage with the strongest mathematical foundation maps directly onto both their heaviest regulatory calculations and parts of their investment process. On the underwriting and capital side, valuing life, health and long-tail property-and-casualty liabilities can require simulations extending decades into the future. Those valuations feed reserving, stress testing, economic-capital calculations and Solvency II coverage ratios, often through layers of nested simulations that become

computationally expensive. Amplitude estimation targets exactly this kind of workload: best-estimate liability valuation, solvency-capital and Value-at-Risk computation and the economic-scenario generation that feeds them. Faster simulation could improve efficiency, but more importantly it could change the cadence of risk management, moving capital and solvency monitoring from a periodic exercise toward more frequent updates and more dynamic steering and hedging. The same techniques carry over to the asset and asset-liability management side. Evaluating a proposed allocation across thousands of economic scenarios, generating those scenarios, and measuring tail and counterparty risk all belong to the same simulation-intensive family that amplitude estimation accelerates.

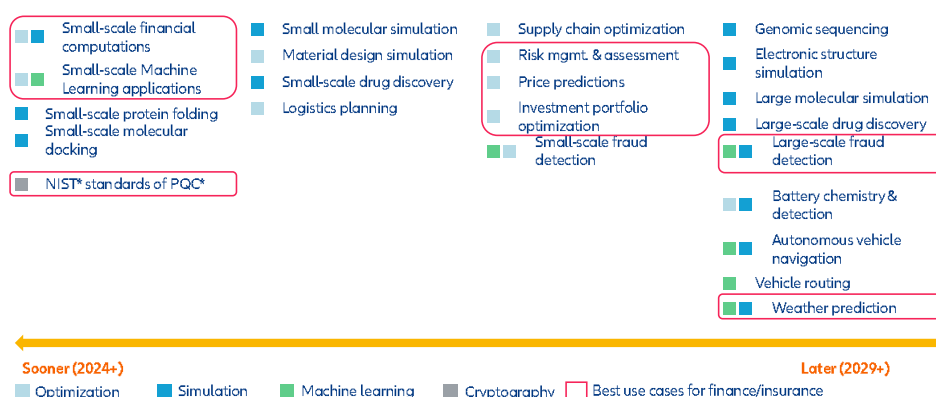
This is also where the wider industry is concentrating its experiments. Goldman Sachs has explored quantum Monte Carlo for derivative pricing; JPMorgan has tested optimization routines for portfolio and collateral problems and Crédit Agricole has piloted hybrid credit-risk models. The honest split is that evaluating an allocation is the part with a proven speedup, whereas choosing the optimal allocation is a combinatorial-optimization problem for which quantum methods have yet to show a robust advantage. The credible opportunity therefore lies first in evaluating allocations faster and with richer scenarios, rather than in expecting a quantum machine to discover the optimal portfolio automatically.

Beyond simulation, the list of proposed applications is long, but the evidence base less so. Corporate and investment banks are testing collateral allocation, balance-sheet optimization and faster pricing systems. Asset managers are exploring portfolio construction, asset modeling and scenario analysis. Payments businesses see potential applications in fraud detection, compliance and, further out, quantum-secure settlement. Retail banks may eventually use quantum or quantum-inspired methods for credit assessment and large-scale decision problems (Figure 2).

At the most experimental end lies quantum money. First proposed by Stephen Wiesner and later developed by Charles Bennett and Gilles Brassard, the concept does not describe a new currency comparable with cryptocurrencies or central-bank digital money, but rather payment tokens encoded in quantum states that cannot be perfectly copied without disturbing them. In principle, quantum mechanics could therefore provide a new way of authenticating value and preventing counterfeiting. Literal quantum banknotes remain technologically remote, however, as they would require fragile quantum states to be stored, transferred and verified reliably, while secure public verification remains an open research problem. More recent experiments have shifted toward quantum payment tokens and one-time cryptograms, using quantum communication to make payment authorization harder to forge without requiring users to preserve a quantum state indefinitely. These approaches remain far from commercial deployment, but they illustrate a broader potential reward.

Most of these opportunities are less certain than the Monte Carlo case. Optimization and machine-learning applications face intense competition from classical solvers that have improved rapidly over the past decade, and a result demonstrated on a small or synthetic problem is not necessarily relevant to an institution operating with noisy data, latency constraints, regulation and legacy infrastructure. The likely economic impact is therefore concentrated in a narrower set of applications than the long lists of potential use cases suggest.

Figure 2: Quantum use cases by expected time to maturity



Sources: Allianz SE Cybersecurity & NextGen IT Think Tank, Arthur D. Little, Allianz Research

Even within that narrower set, value is available today with no quantum hardware advantage. The most dependable gain comes from reformulating the model itself. When you take an actuarial, Asset-Liability-Management or capital model and rewrite it as a QUBO, Ising or tensor form¹, you force an explicit restatement of variables and constraints, surfacing stale constraints and dead risk factors whose corrections feed back into classical production models. Named quantum-inspired solvers are already in commercial use: Multiverse Computing and Crédit Agricole CIB report tensor-network pricing that matches accuracy with fewer parameters and faster training, and Toshiba's Simulated Bifurcation Machine has run low-latency pairs-trading at the Tokyo Stock Exchange. Independent benchmarks find plain simulated annealing comparable in some regimes, and the edge of these solvers comes from specialized hardware rather than a quantum effect. None of this proves a speedup against the moving classical benchmark; the value is real but operational.

In fact, the reward clock has two horizons. A nearer horizon already exists in hybrid quantum-classical workflows and "quantum-inspired" methods that borrow mathematical techniques from quantum computing while running on conventional hardware; platforms such as NVIDIA CUDA-Q couple GPUs with quantum processors and speed up circuit design, simulation and hybrid execution. These gains are operational rather than proof of a quantum speedup, but they lower experimentation costs and help institutions build talent, tooling and credible benchmarks. The far horizon is fault-tolerant quantum advantage: a machine that performs a complete financial task more efficiently than the best classical alternative. Putting a credible number on that long-run reward is not yet possible. The upside is that the reward can therefore arrive the way most technology rewards do: gradually, as capability, tooling and know-how spill over from experimentation into production, with modest gains already available today and a larger advantage possible later. The threat keeps no such schedule. A cryptographic break would not diffuse gradually through the economy, but rather arrive all at once, and its countdown has already begun. It is to that clock that we now turn.

The threat clock

What Shor's algorithm can break extends far beyond any single market. RSA and elliptic-curve schemes secure key exchange, digital signatures, authentication and ownership across the financial system and, in fact, across the wider digital economy that finance depends on. Symmetric cryptography such as AES is also weakened by quantum search but can generally be protected by increasing key lengths; public-key infrastructure requires a more fundamental migration.

The machine does not need to exist for the exposure to exist: adversaries can harvest now and decrypt later.

An adversary can collect encrypted information today and store it for future decryption once suitable hardware arrives, a practice known as "harvest now, decrypt later". For data that must remain confidential for a decade or more, like merger discussions, customer records, trading strategies and health information, the clock has effectively already started. Quantum threat is unusual because part of the future exposure sits in the past: information transmitted years before a cryptographically relevant machine exists may later become readable.

Is Q-Day coming? Expectations of when that machine arrives have moved materially. No general-purpose machine has yet run Shor's algorithm at a cryptographically meaningful scale, but recent studies have sharply reduced estimates of the physical and logical resources required to attack commonly used schemes, and expert opinion has shifted accordingly: The Global Risk Institute's 2025 *Quantum Threat Timeline Report*, answered by 26 experts, assigns a 28–49% probability to a cryptographically relevant quantum computer within ten years, depending on the definition used. The range is wide and should be treated as expert judgment rather than an objective probability, but it is sufficient to make early migration to quantum-safe infrastructure economically rational. Cryptographers formalize the logic as Mosca's inequality: whenever the time needed to migrate plus the required secrecy lifetime of the data exceeds the time until a capable machine arrives, the rational moment to start is already now. The cost of starting too early is a phased technology program; the cost of starting too late could be the loss of trust in critical infrastructure.

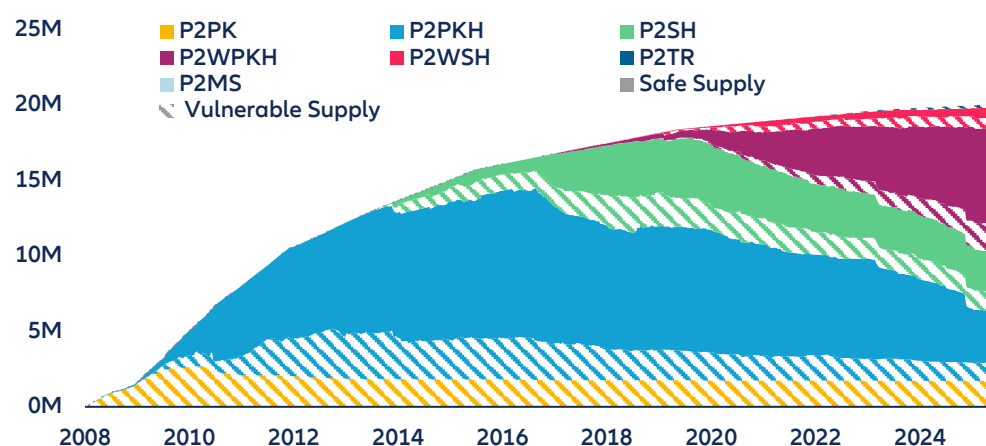
¹ Mathematical forms originally designed for quantum or quantum-inspired solvers, where QUBO stands for Quadratic Unconstrained Binary Optimization, essentially a way of expressing a problem as binary yes/no choices with trade-offs between them.

The defensive mathematics is already settled: In August 2024, the US National Institute of Standards and Technology (NIST) finalized the first post-quantum cryptographic standards, including ML-KEM for key establishment and ML-DSA for digital signatures. Regulators are attaching dates to the transition: NIST plans to deprecate today's vulnerable public-key algorithms after 2030 and disallow them after 2035, turning the migration deadline into a compliance fact rather than a physics forecast. The key uncertainty is therefore no longer whether replacement algorithms will exist, but whether institutions can identify, test and replace their cryptographic dependencies before the threat becomes operational. The new algorithms carry real engineering costs. Post-quantum public keys, ciphertexts and signatures are often larger than their current counterparts, creating challenges for bandwidth, storage, processing and constrained devices. When the Eurosystem and the BIS tested post-quantum cryptography in Project Leap, message sizes expanded and verification times increased. The increase was manageable in many applications, but material in high-volume or latency-sensitive infrastructure. Industry migration has begun in parallel: JPMorgan has operated a production quantum-secured, crypto-agile network since 2024.

The lesson is organizational rather than scientific. Institutions must inventory every system that relies on cryptography, identify data requiring long-term confidentiality, rank systems by risk and migration difficulty, test vendor readiness and build crypto-agility so that algorithms can be replaced without redesigning the entire application. This resembles a multi-year rebuild of trust infrastructure, not a conventional software patch.

Public blockchains could show what a cryptographic migration looks like when no one is in charge of the migration. Bitcoin provides the clearest illustration. By spring 2026, roughly 6mn Bitcoin, or around 30% of total supply, worth roughly USD400bn at current prices, sat behind public keys already visible on-chain. Some of that exposure is operational and can be mitigated by moving funds to fresh addresses. Some is structural. Early pay-to-public-key outputs and a large share of Satoshi-era coins reveal public keys by design, while the corresponding private keys are widely believed to be lost. The chart below (Figure 3) illustrates how that vulnerable supply has evolved across script types over time. The precise number is less important than the principle: a meaningful share of the world's largest digital asset already sits behind cryptography that a future quantum computer could challenge.

Figure 3: Bitcoin supply by script type and vulnerability



Sources: Google Quantum AI, Allianz Research. Note: Labels refer to Bitcoin address types, from the oldest (P2PK, 2009–11) to the newest (P2TR "Taproot", since 2021). What matters is the split: safe supply (solid areas) are coins whose cryptographic key is still hidden and thus cannot be attacked; vulnerable supply (striped areas) are coins whose key is already visible on the blockchain and could in principle be stolen by a powerful future quantum computer. Glossary of the legend: P2PK = Pay-to-Public-Key (key always visible); P2PKH = Pay-to-Public-Key-Hash; P2SH = Pay-to-Script-Hash; P2WPKH / P2WSH = SegWit formats; P2TR = Pay-to-Taproot (key visible by design); P2MS = bare multisig (keys visible)

Ethereum broadens the issue from digital assets to digital infrastructure. The exposure extends beyond user balances to administrative keys controlling stablecoins, tokenized assets, bridges and layer-two networks. As tokenization expands, the same cryptography that secures blockchains will increasingly secure representations of real world financial assets. What begins as a cryptocurrency issue therefore becomes a custody, settlement and investor protection issue for mainstream finance. Industry projections envision tokenized real world assets reaching trillions of dollars over the coming decade; those assets would inherit the same cryptographic vulnerabilities as the networks on which they reside.

Systemic implications

A quantum-enabled cryptographic systemic break would differ from most cyber events for which financial institutions currently plan. Most attacks affect a particular firm, service or network and can eventually be contained and restored. A break in a widely used cryptographic method could spread much further because banks, payment systems and market infrastructures rely on the same small set of standards.

The risk could extend well above the disclosure of confidential information. If public keys and digital signatures could no longer be trusted, an institution might be unable to establish whether a certificate is genuine, whether a message came from the stated counterparty or whether a payment instruction was authorized. An attacker could potentially impersonate a customer or institution, alter instructions or distribute malicious software under an apparently valid signature. The problem would then move quickly from data security to the integrity of payments, settlement and ownership records.

The systemic channel is confidence. Central banks and financial institutions are connected through common payment, messaging and settlement infrastructures, all of which depend on reliable authentication. A problem at one institution would not automatically bring down the system, but uncertainty over whether messages and signatures remained genuine could cause counterparties to delay transactions or suspend activity while the breach was investigated. The stability of the financial system is fundamentally dependent on cryptographic security. Maintaining trust and confidence is therefore central to the response, not merely recovering the data or systems initially affected.

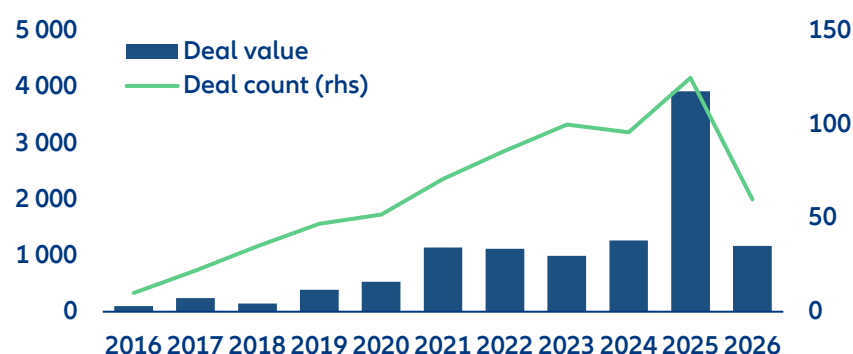
The available loss estimates, though not comparable with each other, point in the same direction. The BIS puts a restrained probability-weighted figure on the danger: financial-system losses equivalent to roughly 0.1% of GDP within five years of a quantum-enabled attack, rising above 1% over a 15- to 20-year horizon, with potentially larger effects in emerging markets. The Hudson Institute examines a more severe tail scenario, modelling a quantum-enabled cyberattack on the Fedwire interbank payment system using a two-stage approach: first calibrating the payment-network contagion effects of an attack on a systemically important bank (drawing on Federal Reserve researchers' Fedwire impairment simulations), then feeding that shock into Oxford Economics' Global Economic Model to trace the macroeconomic path. Their central estimate is a 10–17% decline in US real GDP, equivalent to USD 2.0–3.3trn in GDP at risk (3% of World GDP). Hudson separately offers a cruder, earlier estimate — USD 726bn–1.95trn — for a single-day disruption to one of the five largest US banks' Fedwire access, extrapolated from prior work on quantum threats to the power grid and cryptocurrencies rather than from the full macro model; Hudson itself flags this figure as preliminary and imprecise. The assumptions and methodologies differ substantially across and even within these studies, so the numbers should not be read as forecasts or combined into a single estimate. Their common message is that a failure of financial cryptography would be a macro event with sizeable spillover in the economy, not an IT incident.

Follow the money: capability is accelerating faster than readiness

If the systemic stakes are this high, the natural question is whether investment is flowing toward the machines, the defenses or both. The two are not equally visible: capability spending shows up cleanly in venture and public market data, while defending from the threat is happening behind closed doors. It concerns financial institutions' internal IT budgets, cybersecurity vendors and infrastructure-replacement programs, rather than in flows classified as quantum investment. So, the answer the data can give is necessarily partial and biased. Capital is flowing rapidly into quantum capability: Quantum-technology start-ups raised an estimated USD12.6bn in 2025, with more than

90% directed toward computing rather than sensing or communications. Within that total, venture investors deployed roughly USD3.9bn across around 125 rounds, more than triple the previous year (Figure 4).

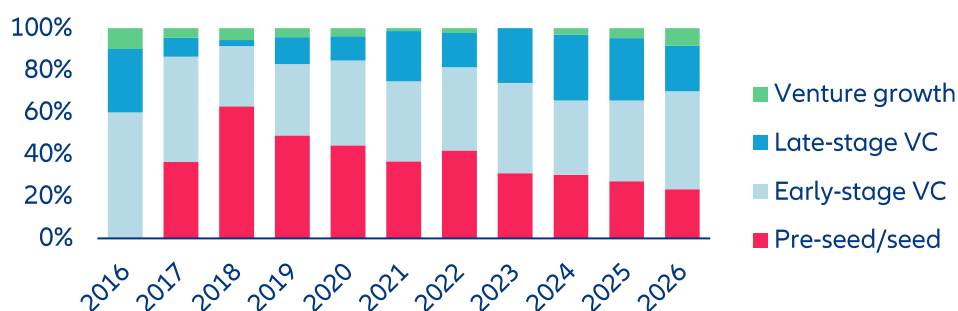
Figure 4: Quantum-computing VC deal activity by quarter, deal value (USD mn) and deal count



Sources: PitchBook, Allianz Research. Note: 2026 is YTD values.

The composition of that capital has changed even faster than its size. As recently as 2024, late-stage and venture-growth rounds were a rounding error; in 2025, venture growth alone jumped from around 1% to more than 30% of deal value, the largest single-year stage shift in the dataset, while mega-rounds of USD25mn and above quadrupled to roughly USD3.5bn. The investors writing those checks are no longer specialist quantum funds taking science risk but crossover institutions, strategic corporate balance sheets and sovereign wealth, some of the same names that anchor late-stage AI and semiconductor rounds. In capital markets terms, quantum has graduated from venture curiosity to institutional allocation. (Figure 5).

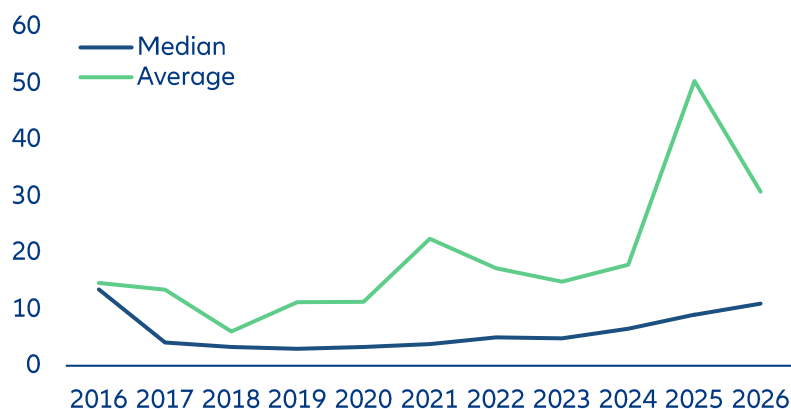
Figure 5 Share of quantum-computing VC deal value by stage



Sources: PitchBook, Allianz Research

The surge in capital has not made the market any broader. The category is governed by an unusually steep power law: the five largest deals account for roughly 27% of all venture value, and the five largest investors for more than 70%. The two largest rounds of the year were struck at pre-money valuations of roughly USD10bn and USD6bn, against a 2025 median pre-money valuation of under USD35mn. That gulf between the average and the median is the signature of a winner-take-most market, and it carries a direct portfolio consequence: the exposure is far easier to obtain than to underwrite. The listed proxies make the same point: pure-play quantum equities trade with betas of 2.5–3x the broad market, and the leading thematic ETF saw assets swell from around USD1bn to nearly USD6bn in 18 months. This is a beta bet on a narrative, not yet a cash-flow story (Figure 6).

Figure 6: Median and average quantum computing VC deal value, USD mn

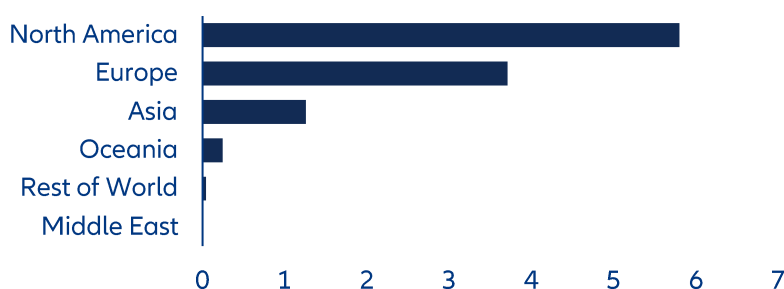


Sources: PitchBook, Allianz Research

The harder truth sits on the other side of the holding period. Quantum is deep-technology investing in its purest form, because it takes a decade or more from laboratory to liquidity and that horizon has so far left little room to exit: public-market debuts have generally underwhelmed, the underlying businesses are too young and too dependent on talent and intellectual property to suit a leveraged buyout and realized proceeds were close to nil as recently as 2024. The partial reopening in early 2026, a single quarter that produced more exit value than the previous three combined, reads more like a release of pent-up pressure than the arrival of a functioning exit market. The real conclusion is about duration rather than returns: the reward clock asks investors for a ten- to fifteen-year lockup, while the threat clock may strike inside a decade.

The narrowness is geographic as well as financial. Since the category took shape, roughly five of every six venture dollars have been committed in North America and Western Europe, while much of Asia's effort runs through state programs that never surface in venture statistics. Even within the West the split is lopsided: Europe attracts only around 5% of global private quantum investment, against more than half for the US, which is positioned to convert its advantages in capital markets, semiconductors and cloud infrastructure into technological leadership. The companies raising the money are also older than they used to be, with the median age at financing nearly doubling over the past decade from around two to four years. The picture is of a market that is deep but not wide: a concentrated pool of large, late and geographically clustered bets whose returns will be decided by a handful of outcomes rather than by the average (Figure 7).

Figure 7: Quantum-computing VC deal value by region, 2016-2026 (USD Bn)



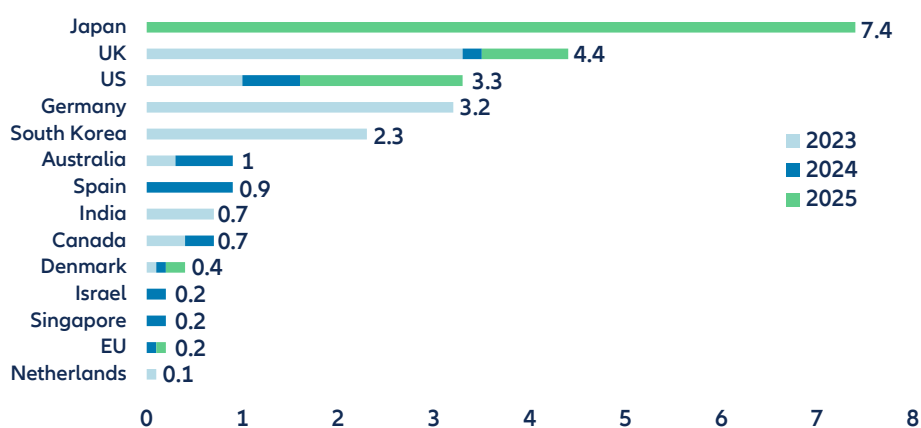
Sources: PitchBook, Allianz Research

Most revealing of all is where the money goes. Of every venture dollar committed to quantum, around 97% has gone to the technology stack – the qubits, the control systems, the platform layer – and less than 1% to financial-services applications. Private capital, in other words, is funding the machine's capability, not the applications that would benefit finance directly. The investor base financing that capability is, to a large extent, the same base that ultimately relies on the cryptographic foundations of the financial system, so the industry has a clear and shared

interest in seeing resilience advance in step with capability. For now, investment in building the technology is running well ahead of investment in preparing for its consequences.

Governments remain important sponsors, and in Washington, policy now addresses both clocks at once. Around USD11.4bn of new public quantum-funding commitments were announced in 2025, led by Japan, the US, the UK and Spain, while earlier multi-year programs in Germany, South Korea and China remain substantial (Figure 8). The US executive order *Ushering in the Next Frontier of Quantum Innovation* seeks to accelerate computing, sensing and networking; strengthen supply chains and workforce development; expand public-private partnerships and deploy a scientific quantum computer at a Department of Energy facility. A parallel order, *Securing the Nation Against Advanced Cryptographic Attacks*, directs agencies to appoint migration leads, inventory cryptographic assets, pilot post-quantum systems and move high-impact federal systems toward post-quantum key establishment and signatures around the turn of the decade. The US is thus addressing capability and defense through a single, coordinated federal instrument.

Figure 8: Government announced commitments on quantum by year, USDbn



Sources: McKinsey, Allianz Research. Note: Figures reflect press-search coverage of announced government commitments only, Jan 2023– Dec 2025; France's national quantum strategy (around €1.8B, Plan Quantique) falls outside this window as it was announced in 2021, and additional country-level commitments may be missing where they run through EU-level programs (e.g., Quantum Flagship) or lacked sufficient press coverage/breakdown to meet inclusion criteria. Figures should be read as directional signals of announcement activity, not a comprehensive accounting of national quantum investment.

Europe is not standing still, but its response is structurally more diffuse. The European Commission's Quantum Europe Strategy, published in July 2025, targets European leadership by 2030, and a proposed Quantum Act, expected in 2026, would try to turn that ambition into industrial policy and coordinated funding. On the defensive clock, at least, Europe has set dates: The EU's coordinated post-quantum roadmap calls on member states to adopt national transition strategies by end-2026 and for critical infrastructure, including finance, to complete migration by 2030. On the capability side, however, the difficulty is fragmentation, which the strategy itself identifies as the central obstacle: European support is spread across Horizon Europe, EuroHPC, the Chips Act, national strategies and a dozen other instruments, without a comparably deep single market for scale-up capital. Where Washington can act through parallel executive orders, Brussels must first coordinate member states and programs. After artificial intelligence, quantum technologies risk becoming another arena in which the US converts a unified market and deeper capital into a durable lead, while Europe legislates and regulates a technology it does not yet own.

The larger gap, however, is not between regions but inside institutions. Only around 35% of large organizations have completed a comprehensive inventory of cryptographic assets, and a smaller share combines executive governance, budgets and a credible migration roadmap. Post-quantum migration is not principally a quantum-research project. It is a large IT and engineering program involving inventory, procurement, architecture, testing, certificate management, hardware replacement and coordination with vendors and counterparties. And the gap runs on both sides of the ledger: just as few institutions have a credible migration roadmap, few have moved beyond

scattered pilots to the systematic talent, benchmarks and integration capacity needed to capture the reward when it arrives. Prepared institutions are the exception on either clock.

Its binding constraints are therefore budget, project capacity and executive attention, the same resources being absorbed by artificial intelligence. The competition is concrete and largely fought within the same IT budget line: the engineers, cloud spend and board time that could fund a cryptographic-migration program are the ones being directed toward AI, which offers more visible and immediate returns. Post-quantum defense risks losing that internal contest not because the threat is unknown, but because its payoff is an avoided loss rather than new revenue, and its deadline remains uncertain.

None of this weakens the opportunity. The investment data show genuine conviction, growing institutional participation and the early development of an industrial ecosystem, and the Monte Carlo case gives insurers in particular a legitimate reason to experiment on both sides of the balance sheet. But the discipline applied to capturing the upside should be matched by equal discipline on timing and resilience, because the two clocks reward different behavior. Capturing the reward is a wager on when quantum beats a moving classical benchmark, and it can afford to wait for evidence. Managing the threat cannot: Harvest-now-decrypt-later means today's traffic is already exposed to tomorrow's machine, and migration itself takes years. That is why quantum preparedness is best understood as an IT-resourcing decision rather than a bet on an arrival date: Crypto-agile architectures, prioritized inventories, migration roadmaps and tested fallbacks pay off across almost every timeline, while waiting pays off only in the one where the machine never comes. What to watch from here: a credible demonstration of factoring at cryptographic scale, logical-qubit counts scaling past error-correction thresholds, the first mandated live post-quantum payment rail, credible business advantages to the financial sector, enforcement of the 2030 depreciation deadlines and the annual shift in the expert threat timeline. Institutions do not need to know which clock will strike first to know that preparing for only one is the wrong strategy.

These assessments are, as always, subject to the disclaimer provided below.

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